

Power BI Assignment – E-Commerce

Microsoft Power BI

From Raw Data to Insight

E-Commerce Sales, Customer, Product and Fulfilment Analysis

Scenario: Question 1 — E-Commerce Company

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Course: Microsoft Power BI

Date: 1 October 2026

Dashboard link: https://livemecedu-my.sharepoint.com/:u:/g/personal/23f24281_mec_edu_om/IQAb9ZKxRp9aS5DzNEIaRRh3Ade6ZbljWGuN0I6yOX0AjAc?e=Ak2iTn

1. Introduction

The report discusses the performance of the online retailer based on the provided E-Commerce Dataset (Question 1). The target audience is the company's management team that needs to have the information on their performance dashboard for the weekly trading meetings. This report answers the following questions: the most profitable categories and sales channel; the most valuable customer segment and geographical area; impact of discounts, refunds, and delivery status on performance; and products that need attention. The dataset contains four sheets: Ecom_Orders (80 orders, January–July 2025), Ecom_Customers (12 customers), Ecom_Products (10 products) and Ecom>Returns (18 returns).

2. Data Cleaning in Power Query

The sheets were imported as queries and re-named Orders, Customers, Products and Returns. The most significant Applied Steps (Figure 1) were:

Data type: header promotion and data types changed to Date for Dates, integers for Quantities, decimal for Revenue and Discount, and text for IDs.

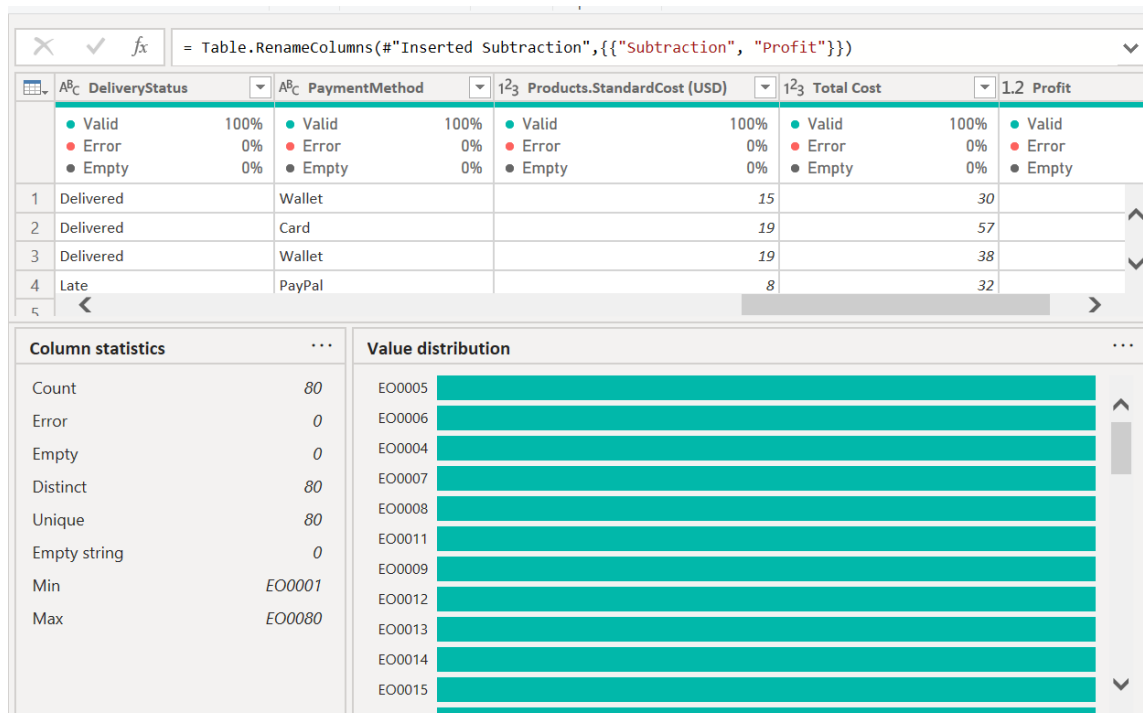
Data cleaning: blank rows and error rows deletion and de-duplication based on OrderID column. There were no duplicates found and hence all the 80 orders were retained.

Renaming: "UnitPrice (USD)" field was renamed to Unit Price and "Revenue (USD)" field was renamed to Revenue.

Calculation of derived fields: Orders was joined with Products to extract the StandardCost field. Total Cost = Quantity * StandardCost, Profit = Revenue - Total Cost.

Validation: each CustomerID, ProductID and returned OrderID value corresponds to the corresponding parent row and

Channel, Segment, Region and Category fields were consistent.



Move	Split Column	Format	Extract	Parse
Convert to List				
SelectRows("#"Changed Type", each no				
CustomerName	Segment			
Ava Patel	Consumer			
Ben Carter	Corporate			
Chloe Martin	Consumer			
Diego Ruiz	Home Office			
Emma Wilson	Consumer			
Fatima Noor	Corporate			
Grace Kim	Consumer			
Henry Lee	Home Office			
Ivy Chen	Consumer			
Jon Bell	Corporate			
Kara Singh	Consumer			
Leo Brown	Home Office			

- lowercase
- UPPERCASE
- Capitalize Each Word
- Trim
- Clean
- Add Prefix
- Add Suffix

1.2 Total Cost	1.2 Profit
30	32
57	51
38	34
32	40
24	27.3
11	13
44	52
33	39
38	30
152	92.8
26	30
13	13.6
84	74
69	57
52	60
42	37
46	38
30	32.7
116	100
23	19
20	24

Place Values ▾	Unpivot Columns ▾	Split Column ▾	Format ▾	Merge Columns ▾
Column	Move ▾		ABC 123 Extract ▾	
Column	Convert to List		abc Parse ▾	
SelectRows("#Changed Type", each no				
ABC CustomerName	ABC Segme		lowercase	
			UPPERCASE	
			Capitalize Each Word	
			Trim	
			Clean	
			Add Prefix	
			Add Suffix	
Ava Patel	Consumer			
Ben Carter	Corporate			
Chloe Martin	Consumer			
Diego Ruiz	Home Office			South
Emma Wilson	Consumer			West
Fatima Noor	Corporate			North
Grace Kim	Consumer			East
Henry Lee	Home Office			South
Ivy Chen	Consumer			West
Jon Bell	Corporate			North
Kara Singh	Consumer			East
Leo Brown	Home Office			South

3. Data Modeling

The data model uses the principle of star schema design (see Figure 2). “Orders” is the fact table, whose level of granularity is one row per order. Dimension tables include “Customers” and “Products”, while “Returns” is the event table and “Calendar” is the time dimension. Relationships:

Customers[CustomerID] → Orders[CustomerID]: one-to-many, single direction.

Products[ProductID] → Orders[ProductID]: one-to-many, single direction.

Calendar[Date] → Orders[OrderDate]: one-to-many, single direction.

Orders[OrderID] ↔ Returns[OrderID]: bidirectional, so region, category and channel slicers also filter return visuals.

Key measures (DAX):

Total Revenue = SUM(Orders[Revenue])

Profit Margin % = DIVIDE([Profit], [Total Revenue])

Average Order Value = DIVIDE([Total Revenue], [Orders])

Return Rate % = DIVIDE(DISTINCTCOUNT>Returns[OrderID]), [Orders])



4. Dashboard Design

Each page addresses a different business question and contains one dark color scheme, consistent titles, a text box for an explanation, and navigation buttons. The charts contain zero-based axes to avoid distortion in comparison.

Page 1: Executive Overview. Purpose: To demonstrate the health of our sales for the weekly trading meeting. It has KPIs for revenue, orders and profit, revenue line chart, revenue by channel donut chart, revenue by region bar chart, and slicers for channel and region to filter each visual.

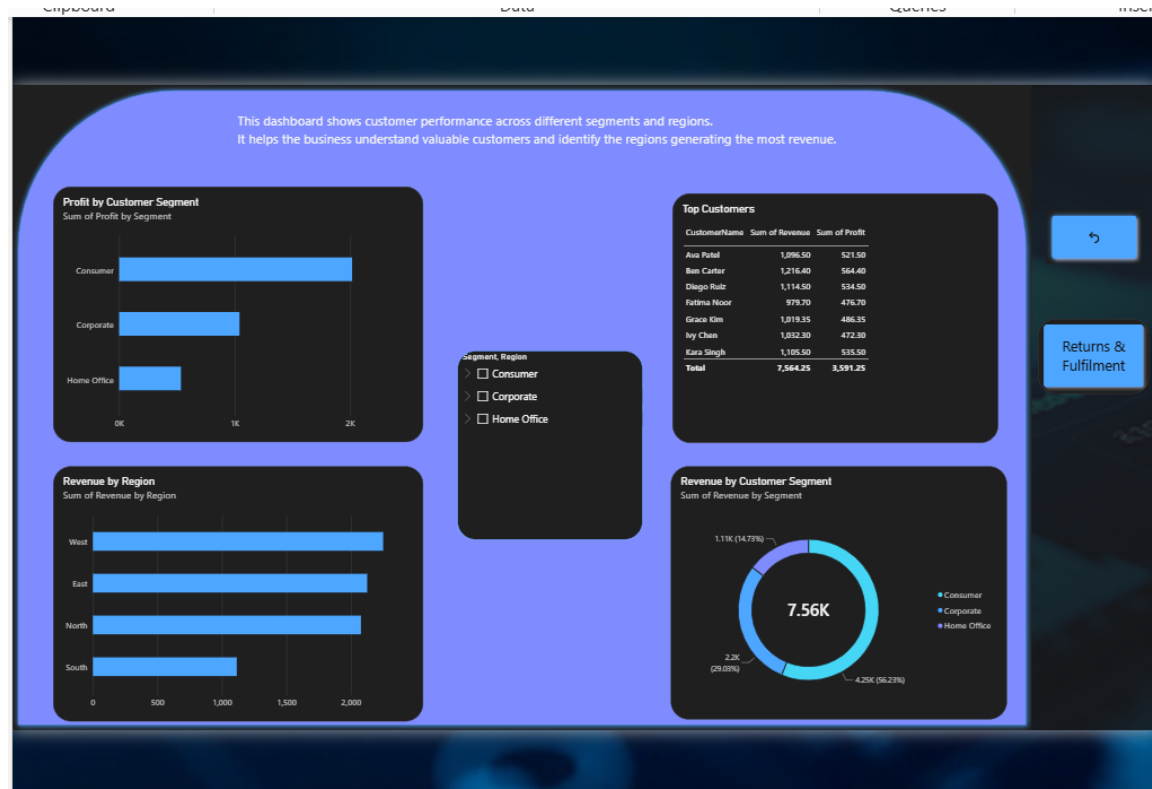


Page 2 - Product Performance.

Objective: determine which products need to be prioritized, reviewed, and promoted. It includes category-wise revenue, product-wise profit, revenue vs profit scatter plot, units by category, discount details table, and category slicer



Page 3 - Customer and Geography. Purpose: Demonstrate who and where the high-value customers are. The page consists of profit per segment, revenue per region, revenue per segment donut, top customers, and slicers for segments and regions.



Page 4 - Returns & Fulfillment. Purpose: demonstrate the impact of returns and fulfillment on performance. Displays the overall number of refunds, returns per month, percentage and number of reasons for refunds, and orders based on delivery status.



5. Interpretation

Overall: revenue was \$7,564 and profit \$3,591 (47.5% margin) from 80 orders; average order value was \$94.55 (Figure 3).

Trend: revenue peaked in February (\$1,622), then fell sharply from May, when monthly orders dropped from 14–15 to 7–8.

Channels and regions: the three channels are almost equal (\$2,475–\$2,576). West leads on revenue (\$2,249) but has the lowest margin (46.1%). South is smallest (\$1,115).

Products: Bags earn the most revenue (32%), led by Everyday Backpack (\$1,272). Home Office has the lowest margin (43.2%); Monitor Stand is weakest (41%) with the highest average discount. Accessories and Hydration exceed 52% margin but sell little (Figure 4).

Customers: Consumers generate 56% of revenue. Ben Carter is the top customer (\$1,216), and value is evenly spread (Figure 5).

Discounts: discounted orders (24%) earn a 42.8% margin versus 48.8% at full price.

Returns and delivery: 18 returns (22.5% of orders) cost \$486 in refunds. Damaged and changed-mind returns are the top reasons. Late orders are 25% of orders but 38% of revenue, so delays put the largest sales at risk (Figure 6). Revenue includes cancelled orders (\$1,411).

6. Final Conclusion

Business remains profitable, but order volume is down about half from May and late delivery issues and returns jeopardize the biggest orders. Recommendations:

Improve fulfillment process by checking carrier and package performance to reduce late deliveries and damaged products that create returns, and watch weekly Return Rate .%

Redistribute sales efforts to focus on Accessories and Hydration categories, reduce Monitor Stand discounts, and conduct West/South campaigns. **Dashboard**

link: [https://livemecedu-](https://livemecedu-my.sharepoint.com/:u:/g/personal/23f24281_mec_edu_om/IQAb9ZKxRp9aS5DzNElaRRh3Ade6ZbljWGuN0I6yOX0AjAc?e=Ak2iTN)

[my.sharepoint.com/:u:/g/personal/23f24281_mec_edu_om/IQAb9ZKxRp9aS5DzNElaRRh3Ade6ZbljWGuN0I6yOX0AjAc?e=Ak2iTN](https://livemecedu-my.sharepoint.com/:u:/g/personal/23f24281_mec_edu_om/IQAb9ZKxRp9aS5DzNElaRRh3Ade6ZbljWGuN0I6yOX0AjAc?e=Ak2iTN) **Sources:** Power BI Assignment Sample Datasets.xlsx (synthetic data). No external sources were used.